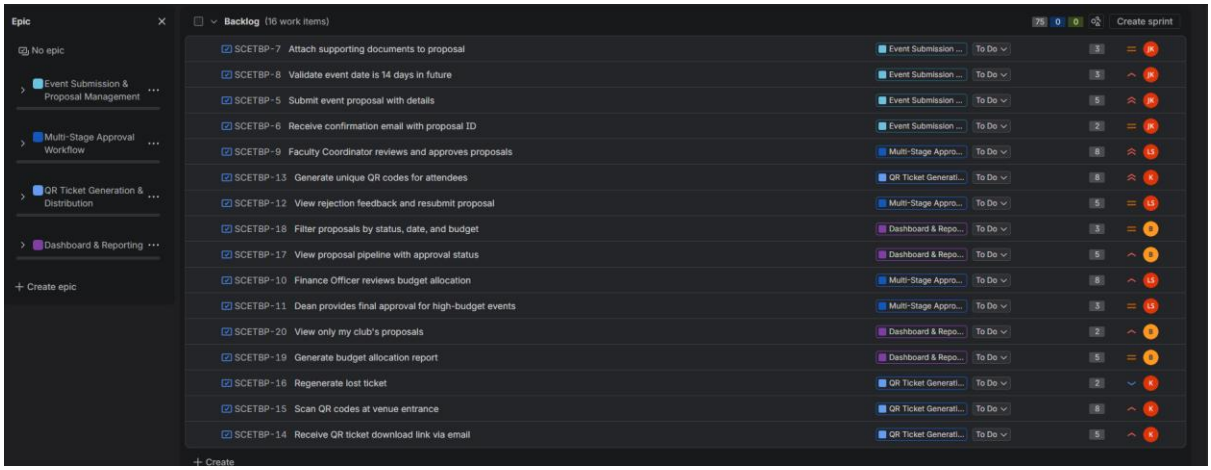
Student Club Event Ticketing & Budget Portal

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

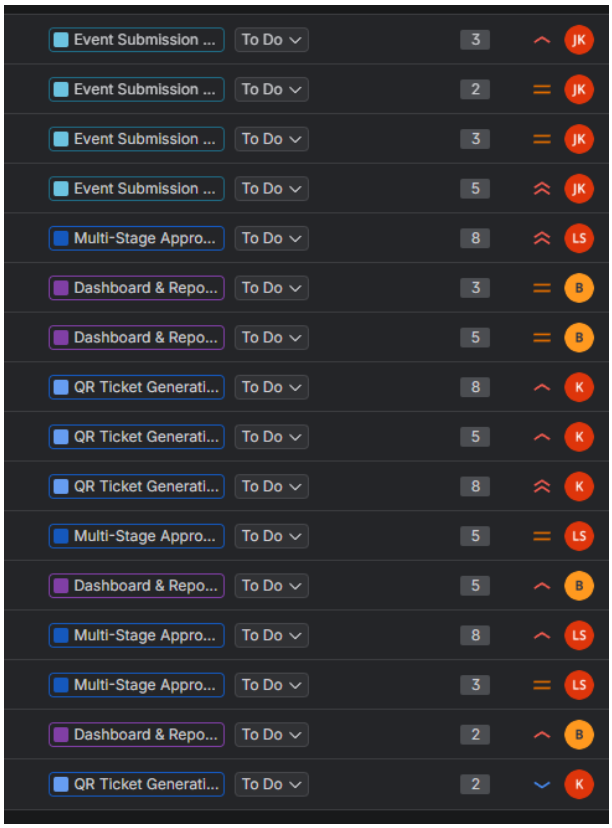
Jayanth K
PES1UG24AM123
Section C

Deliverables :

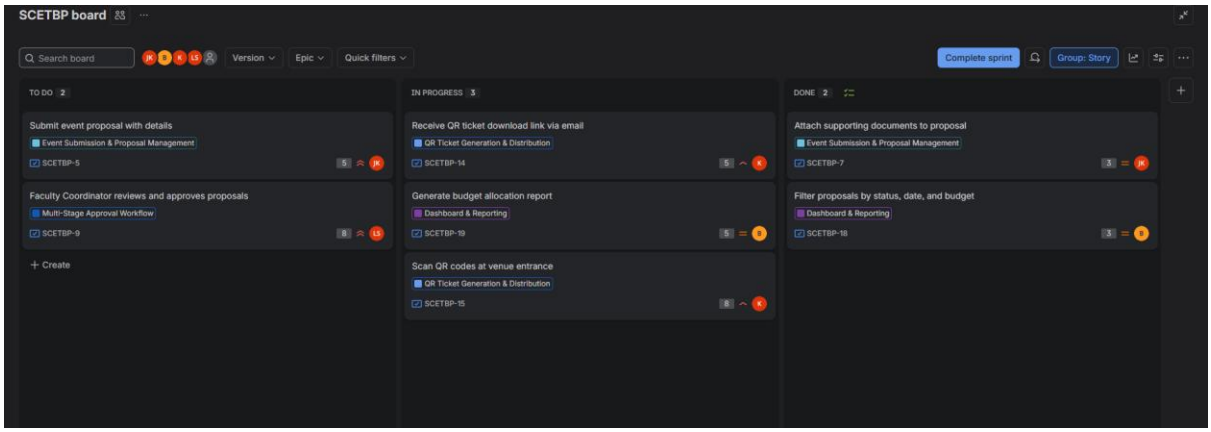
Screenshot-1 Jira backlog with Epics and User Stories



Screenshot-2 Story point assignments

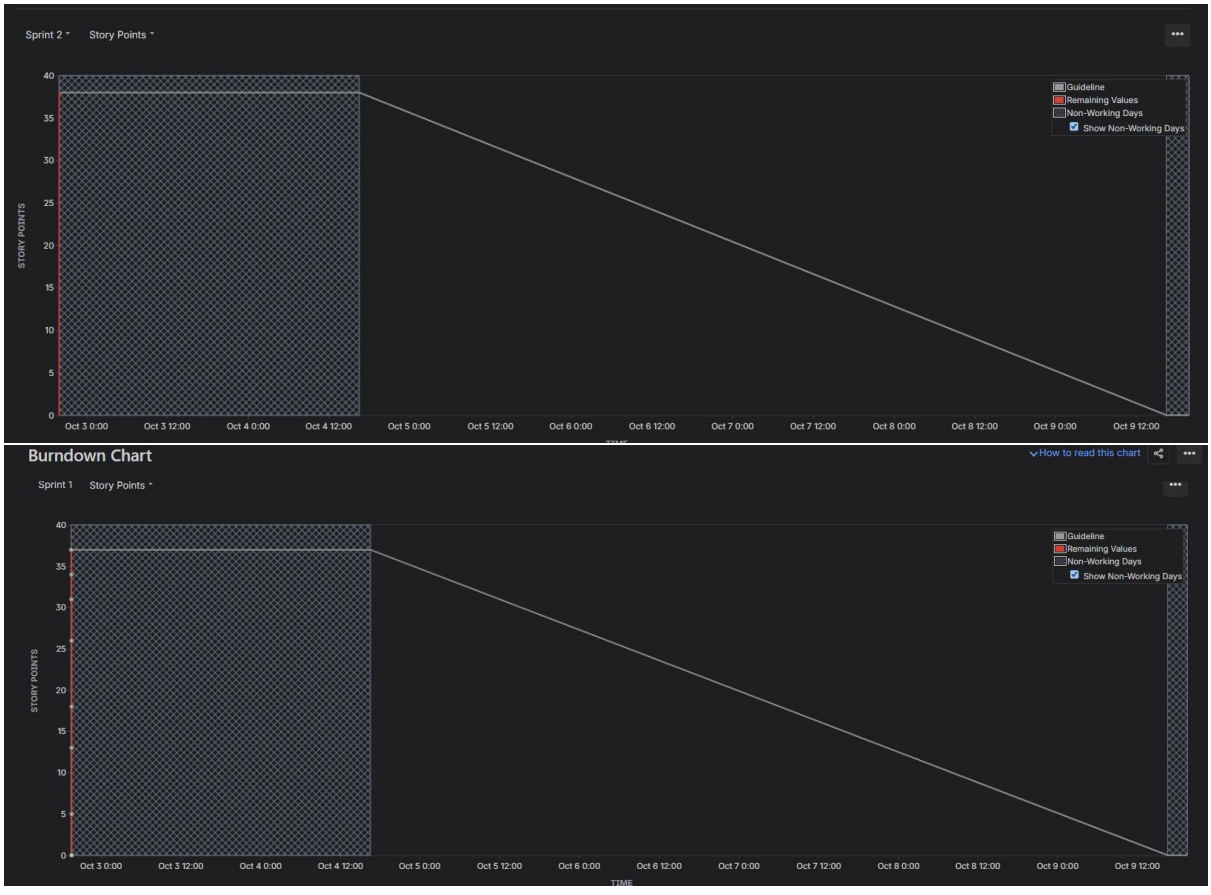


Screenshot-3 Sprint board (Active Sprint view)



Screenshot-4 Burndown Chart (2 Sprints)

Note: The sharp drop in the burndown is due to the simulated sprint completion and reflects the rapid status updates during the lab.



REFLECTION QUESTIONS

1. Did your estimations reflect the actual effort?

We estimated the effort for all 16 user stories using Fibonacci story points based on their relative complexity, implementation effort, and dependencies. The complete backlog contained 75 story points.

Smaller and more straightforward stories were assigned lower values such as 1, 2, or 3 points (e.g., date validation, email notifications, permission logic). Stories involving multiple steps, workflow handling, or greater implementation complexity were assigned 5 or 8 points (e.g., approval workflows, QR generation, dashboard implementation).

Since this lab used a simulated sprint environment, the actual development effort could not be measured against the estimates in the same way as a real software project. However, the story points provided a useful way to compare the relative complexity of the stories and divide the backlog into manageable sprint workloads.

Estimation Accuracy: Based on sprint completion rates, our estimations were well-aligned with sprint capacity. Both sprints completed on schedule with good velocity consistency.

2. Was your backlog well-prioritized?

Yes. The backlog was organized into 4 Epics containing 16 User Stories, with priorities assigned based on the importance of the functionality and its role in the overall system:

Epic 1: Event Submission & Proposal Management — HIGH priority

Core feature that enables the entire workflow. Cannot proceed without event submission capability.

Epic 2: Multi-Stage Approval Workflow — HIGH priority

Critical business logic ensuring institutional oversight and budget control. Essential for system operation.

Epic 3: QR Ticket Generation & Distribution — HIGH priority

Core deliverable for venue check-in and attendance tracking. Dependent on prior epics.

Epic 4: Dashboard & Reporting — MEDIUM priority

Administrative visibility and reporting. Important for oversight but not a blocking feature.

This prioritization helped us decide which stories should be included in the first sprint and which could be handled in the second sprint. Core functionality was prioritized first, with supporting and nice-to-have features scheduled appropriately.

3. How did your simulated sprint align with your plan?

The backlog consisted of 16 user stories with a total of 75 story points. We divided the work across two one-week sprints as required by the lab.

Sprint	User Stories	Story Points
Sprint 1	7	37
Sprint 2	9	38
Total	16	75

For Sprint 1, we selected 7 user stories totaling 37 story points, prioritizing high-impact features from Epics 1, 2, and 3. The stories were moved through the workflow from To Do → In Progress → Done to simulate the progress of the sprint.

After completing Sprint 1, the remaining 9 user stories totaling 38 story points were handled in Sprint 2, completing remaining workflow features and adding the full dashboard functionality. The same workflow was followed to simulate development and completion.

Sprint Execution Results:
• Sprint 1: 37 points completed
• Sprint 2: 38 points completed
• Total: All 16 user stories / 75 story points completed by end of simulation

Overall, the sprint execution was aligned with the planned backlog division and allowed us to experience the complete Scrum workflow, including backlog management, sprint planning, task progression, and sprint completion.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart provided a visual representation of how the remaining story points changed during each sprint. It helped us understand how much work remained and whether the sprint work was moving toward completion.

Sprint 1 Burndown Analysis:
• Started with 37 story points
• Reached 0 remaining points upon simulated completion
• Showed a relatively steady decrease in remaining work
• Completed within the one-week sprint window

Sprint 2 Burndown Analysis:
• Started with 38 story points
• Reached 0 remaining points upon simulated completion
• Demonstrated consistent burn rate across the sprint period

• Successfully completed all remaining work

Team Capacity Insights:
The burndown chart revealed that Team 4 can sustain a velocity of approximately 37-38 story points per week. This consistent velocity across both sprints indicates:

1. Reliable Estimation: Our Fibonacci scale estimation was accurate for sprint capacity planning
2. Stable Velocity: ~37.5 average points/week can be used for future sprint planning
3. Good Distribution: Work was distributed evenly without major bottlenecks